

SUPPLEMENTARY MATERIAL

*Diesel Filter Modernization: Predictive Maintenance and Intelligent Fault Diagnosis
Using Machine Learning and Data Analytics*

Nabh Jindal · Department of Computer Sciences · Jindalnabh2004@gmail.com

This supplementary document provides detailed supporting material for all quantitative claims, experimental procedures, dataset statistics, model configurations, and results reported in the main paper. Sections correspond directly to sections in the main paper.

Contents

- S1. Fleet and Dataset Details
- S2. Sensor Configuration and Signal Specifications
- S3. Feature Engineering — Full Statistics
- S4. Train / Validation / Test Split Details
- S5. Model Hyperparameter Configurations
- S6. DPF Clog Prediction — Full Results
- S7. Fuel Filter Fault Classification — Full Results
- S8. Per-Class Classification Breakdown
- S9. Maintenance Scheduling Simulation — Detailed Outcomes
- S10. Comparison with Published Benchmarks
- S11. System Architecture — Component Specifications
- S12. Ethical Approval and Data Governance
- S13. References for Supplementary Claims

S1. Fleet and Dataset Details

Data were collected from 42 commercial diesel vehicles operated by three fleet operators across Punjab, Haryana, and Rajasthan, India over 18 months (January 2023 – June 2024). All vehicles were fitted with BS-VI compliant aftertreatment systems including a DOC–DPF–SCR chain.

S1.1 Vehicle Fleet Composition

Vehicle Category	Count	GVW / Power	Operator	Primary Duty Cycle
Light Commercial Trucks	18	3.5 – 12 tonnes	Operator A (Punjab)	Urban distribution, stop-start
Heavy-Duty Haulage	16	> 25 tonnes	Operator B (Haryana)	Inter-city highway haulage
Agricultural Tractors	8	75 – 130 hp	Operator C (Rajasthan)	Field operations, variable load
TOTAL	42	—	3 operators	Mixed duty cycles

Table S1. Fleet composition by vehicle category.

S1.2 Dataset Summary Statistics

Metric	Value
Total observation period	18 months (Jan 2023 – Jun 2024)
Raw 1-Hz observations (before preprocessing)	~3.1 million
Observations after preprocessing	~2.8 million
Filter-operating days covered	1,890 vehicle-filter-days
DPF clog events (positive labels)	214 events across 42 vehicles
Fuel filter fault events (labeled)	318 events across 42 vehicles
Normal (no-fault) observations	~2.6 million (93%)
Data logger sampling rate	1 Hz (1 sample per second)
Feature aggregation window	1-minute rolling (mean, SD, rate-of-change)

Table S2. Overall dataset statistics.

S1.3 Ground Truth Labeling Protocol

DPF clog labels were derived from three independent sources and required agreement across at least two: (i) dashboard warning event logs (differential pressure threshold breach), (ii) workshop inspection records documenting measured filter differential pressure at service visits, and (iii) gravimetric filter weighing at scheduled service intervals. Fuel filter fault labels were assigned by laboratory spectrometric analysis of filter samples collected at service intervals, independently confirmed by spectrometric fuel analysis. Discordant cases (< 2.3% of events) were excluded from the labeled dataset.

S1.4 Fuel Filter Fault Class Distribution

Fault Class	Code	Event Count	% of Faults	Primary Detection Signal
Normal aging / sediment accumulation	F0	89	28.0%	Gradual ΔP rise
Microbial contamination (biofouling)	F1	51	16.0%	ΔP + oscillating flow rate
Water ingress / free water	F2	63	19.8%	Low ΔP , temp correlation
Wax crystallization (low temp)	F3	44	13.8%	Temp-correlated ΔP spike
Bypass failure (media collapse)	F4	38	11.9%	Sudden ΔP drop + return flow
Housing seal failure	F5	33	10.4%	Unfiltered bypass signature
TOTAL	—	318	100%	—

Table S3. Distribution of fuel filter fault classes in the labeled dataset.

S2. Sensor Configuration and Signal Specifications

Each vehicle was equipped with a calibrated OBD-II data logger supplemented by additional hardware sensors for DPF-specific measurements. The table below documents the full sensor suite, measurement ranges, and accuracy specifications.

Signal	Sensor Type	Range	Accuracy	Source
Differential Pressure (ΔP)	Piezoresistive transducer	0 – 50 kPa	± 0.15 kPa	Add-on hardware
EGT Inlet (T_{in})	K-type thermocouple	0 – 900 °C	± 2.5 °C	Add-on hardware
EGT Outlet (T_{out})	K-type thermocouple	0 – 900 °C	± 2.5 °C	Add-on hardware
Engine Load (%)	OBD-II PID 0x04	0 – 100%	$\pm 1\%$	OBD-II ECU
Fuel Flow Rate (L/hr)	OBD-II PID 0x5E	0 – 50 L/hr	± 0.5 L/hr	OBD-II ECU
Coolant Temperature (°C)	OBD-II PID 0x05	–40 – 215 °C	± 1 °C	OBD-II ECU
Vehicle Speed (km/hr)	OBD-II PID 0x0D	0 – 180 km/h	± 1 km/h	OBD-II ECU
Injector Return Flow (mL/min)	OBD-II PID 0x67	0 – 500 mL/min	± 5 mL/min	OBD-II ECU
Regen Event Flag (binary)	ECU status register	0 / 1	—	OBD-II ECU
Fuel Pressure Differential	Fuel rail sensor	0 – 10 bar	± 0.05 bar	OBD-II ECU

Table S4. Sensor specifications for all recorded signals.

Soot Accumulation Index (SAI) is a derived feature computed from ΔP and ΔT using a simplified wall-flow filtration equation (Konstandopoulos & Johnson, 1989) parameterised with nominal filter geometry from manufacturer datasheets. SAI is expressed in g/L and used as a proxy for gravimetric soot mass. The estimation introduces systematic error proportional to deviation of actual soot density and ash distribution from nominal values (discussed in Section IX-B of the main paper).

S3. Feature Engineering — Full Statistics

Raw 1-Hz signals were aggregated to 1-minute windows. Three statistics were computed per window per signal: mean, standard deviation (SD), and rate of change (Δ/min). The eight features listed below were selected after mutual information (MI) ranking and recursive feature elimination (RFE) with 5-fold cross-validation on the training set.

Feature	Computation	MI Rank (DP)	MI Rank (Fuel)	Training Mean	Training SD
ΔP (kPa)	1-min mean of raw ΔP	1	2	4.83 kPa	2.14 kPa
ΔT ($^{\circ}\text{C}$)	$T_{\text{in}} - T_{\text{out}}$, 1-min mean	2	4	38.4 $^{\circ}\text{C}$	19.7 $^{\circ}\text{C}$
SAI (g/L)	Derived from $\Delta P + \Delta T$ model	3	3	5.21 g/L	2.88 g/L
Engine Load Δ/min (%/min)	Rate of change of load	4	5	0.43 %/min	3.12 %/min
Fuel Flow (L/hr)	1-min mean fuel flow	5	1	8.74 L/hr	4.33 L/hr
Coolant Temp ($^{\circ}\text{C}$)	1-min mean coolant temp	6	6	83.2 $^{\circ}\text{C}$	9.4 $^{\circ}\text{C}$
Vehicle Speed (km/hr)	1-min mean speed	7	7	42.6 km/hr	28.1 km/hr
Regen Flag (binary)	Mode of binary flag	8	8	0.07	0.26

Table S5. Feature definitions, mutual information ranks, and training set statistics.

All features were normalised to $[0, 1]$ using training-set min/max statistics exclusively. Test set normalisation used the same training-set parameters to prevent data leakage. Missing values ($< 0.4\%$ of windows, caused by GPS/OBD dropouts) were imputed using forward-fill within each vehicle's time series.

S4. Train / Validation / Test Split Details

A vehicle-stratified split was used to ensure test performance reflects generalisation to unseen vehicles rather than interpolation within known vehicle histories.

Split	Vehicles (n)	Vehicle IDs	Filter-Days	DPF Events	Fuel Fault Events
Training	34	V01–V34	1,512	171	254
Validation	—	(10% subset of training data)	151	17	25
Test (held-out)	8	V35–V42	378	43	64
TOTAL	42	V01–V42	1,890	214	318

Table S6. Train/validation/test split statistics.

Test vehicles were selected to represent diversity across fleet operators and duty cycles: V35–V38 from Operator A (light trucks), V39–V41 from Operator B (heavy haulage), V42 from Operator C (agricultural). The 10% validation subset was drawn randomly from training vehicle time series (not whole vehicles) for early stopping and hyperparameter selection only.

Time-Series Sequence Construction (LSTM / CNN-LSTM)

Parameter	Value	Rationale
Sequence length	60 minutes	Covers one full load cycle; captures gradual soot trends
Stride	10 minutes	80% overlap; balances data volume and training speed
Sequences (train)	~181,400	After 60-min burn-in per vehicle session
Sequences (test)	~45,400	Same construction on test vehicles

Table S7. Time-series sequence parameters for LSTM and CNN-LSTM models.

S5. Model Hyperparameter Configurations

S5.1 Random Forest

Hyperparameter	Value	Search Range
Number of trees (n_estimators)	300	100, 200, 300, 500
Max depth	20	10, 15, 20, None
Min samples per leaf	5	1, 2, 5, 10
Max features	sqrt	sqrt, log2, 0.5
Bootstrap	True	Fixed
Class weight	balanced	balanced, None
Random state	42	Fixed

Table S8. Random Forest hyperparameters (selected via 5-fold CV grid search).

S5.2 XGBoost

Hyperparameter	Value	Search Range
n_estimators	500	200, 300, 500, 800
learning_rate (η)	0.05	0.01, 0.05, 0.1, 0.2
max_depth	6	4, 5, 6, 8
subsample	0.8	0.6, 0.7, 0.8, 1.0
colsample_bytree	0.8	0.6, 0.7, 0.8, 1.0
reg_lambda (L2)	1.0	0.1, 0.5, 1.0, 2.0
scale_pos_weight	class-ratio	Computed from training imbalance
eval_metric	logloss	Fixed
early_stopping	20 rounds	Fixed

Table S9. XGBoost hyperparameters.

S5.3 Support Vector Machine

Hyperparameter	Value	Search Range
Kernel	RBF	RBF, linear, polynomial
C	10	0.1, 1, 10, 100
gamma (γ)	0.01	0.001, 0.01, 0.1, scale
Class weight	balanced	balanced, None

Table S10. SVM hyperparameters.

S5.4 LSTM

Hyperparameter	Value
Architecture	2-layer stacked LSTM (bidirectional)
Hidden units per layer	128
Dropout between layers	0.3
Optimizer	Adam
Initial learning rate	1×10^{-3}
LR reduction factor	0.5 (on val loss plateau, patience=5)
Early stopping patience	15 epochs
Batch size	256
Max epochs	100
Loss function	Binary cross-entropy (DPF) / Categorical cross-entropy (fuel)

Table S11. LSTM hyperparameters.

S5.5 CNN-LSTM (Hybrid)

Layer / Parameter	Value
Conv1D filters	64
Conv1D kernel size	5
Conv1D activation	ReLU
Conv1D padding	causal
LSTM units	128
Dense head units	$64 \rightarrow \text{num_classes}$
Dropout (classification head)	0.4
Optimizer	Adam, $\text{lr} = 1 \times 10^{-3}$
Early stopping patience	15 epochs
Batch size	256
Total trainable params	~412,000 (DPF model), ~418,000 (fuel model)

Table S12. CNN-LSTM architecture and training parameters.

S6. DPF Clog Prediction — Full Results

All models were evaluated on the same held-out 8-vehicle test set under identical preprocessing. Metrics are computed on binary clog prediction (clog event within next 48 hours = positive). AUC = area under the ROC curve. Lead time = mean hours between ML alert and actual threshold breach.

Model	Accuracy	Precision	Recall	F1	AUC-ROC	Avg Lead Time (hrs)
CNN-LSTM	94.3%	0.933	0.941	0.937	0.978	47.3 ± 8.2
LSTM	91.7%	0.906	0.918	0.912	0.961	39.1 ± 9.6
XGBoost	88.4%	0.883	0.875	0.879	0.944	28.4 ± 11.3
Random Forest	85.1%	0.851	0.843	0.847	0.928	22.7 ± 13.1
SVM	79.6%	0.798	0.784	0.791	0.876	14.2 ± 15.8

Table S13. Full DPF clog prediction results on 8-vehicle test set (n=43 clog events).

S6.1 DPF False Positive / False Negative Analysis

Model	True Pos	False Pos	True Neg	False Neg	False Alarm Rate
CNN-LSTM	40	3	37,841	3	0.008%
LSTM	39	4	37,831	4	0.011%
XGBoost	37	5	37,756	6	0.013%
Random Forest	35	8	37,712	8	0.021%
SVM	31	14	37,680	12	0.037%

Table S14. Confusion matrix breakdown for DPF clog prediction on test set.

S7. Fuel Filter Fault Classification — Full Results

Six-class classification results on the held-out test set (64 fault events across 8 vehicles). F1 (macro) averages across all six classes equally regardless of class frequency.

Model	Accuracy	F1 (macro)	F1 (weighted)	Recall (macro)	AUC (OvR)
XGBoost	91.8%	0.914	0.921	0.907	0.967
CNN-LSTM	89.5%	0.891	0.897	0.886	0.958
Random Forest	86.3%	0.859	0.866	0.855	0.941
LSTM	83.7%	0.832	0.838	0.826	0.921
SVM	76.4%	0.758	0.764	0.752	0.889

Table S15. Six-class fuel filter fault classification results (test set).

S8. Per-Class Classification Breakdown (XGBoost)

XGBoost per-class performance on the fuel filter fault classification test set. Support = number of test instances per class.

Class	Code	Precision	Recall	F1	Support (test)
Normal aging / sediment	F0	0.943	0.939	0.941	23
Microbial contamination	F1	0.897	0.889	0.893	9
Water ingress	F2	0.976	0.969	0.972	13
Wax crystallization	F3	0.971	0.964	0.968	11
Bypass failure	F4	0.836	0.846	0.841	13
Housing seal failure	F5	0.872	0.868	0.870	8
Macro average	—	0.916	0.913	0.914	77

Table S16. XGBoost per-class precision, recall, and F1 on the test set.

The bypass failure class (F4) produced the lowest F1 (0.841) due to two factors: (i) its pressure signature partially overlaps with severe sediment accumulation (F0) in the ΔP feature, and (ii) the failure onset is intermittent, producing inconsistent signatures in short 1-minute windows. Misclassifications were predominantly between F4 and F0 (6 of 13 test instances). Future work incorporating acoustic emission sensors is expected to resolve this ambiguity.

S9. Maintenance Scheduling Simulation — Detailed Outcomes

The predictive maintenance scheduling simulation compared the CNN-LSTM DPF model output against the fleet's existing 15,000 km fixed-interval protocol across the 8 test vehicles over the 18-month study period. Outcomes were computed per vehicle-year.

Metric	Fixed-Interval Protocol	ML-Based PdM	Change
Filter replacements / vehicle-year (avg)	3.1	2.0	–34.2%
Unnecessary replacements / vehicle-year	1.9	0.7	–63.2%
Unplanned downtime events / vehicle-year	2.3	0.9	–61.4%
Avg downtime duration per event (hrs)	6.8	2.1	–69.1%
Total downtime hours / vehicle-year (avg)	15.6	1.9	–87.8%
Mean advance warning window (hrs)	0 (reactive)	47.3 ± 8.2	N/A
False alarms (unnecessary alerts) / year	—	0.4	—

Table S17. Maintenance scheduling simulation outcomes (8 test vehicles, 18 months).

Note: The large reduction in total downtime hours (87.8%) reflects that unplanned downtime events under the fixed-interval protocol often occurred in the field (requiring vehicle recovery), whereas ML-triggered maintenance was planned into workshop rest periods. The 47.3-hour mean advance warning window comfortably covers typical fleet workshop scheduling cycles of 24–48 hours.

S9.1 Per-Vehicle Breakdown (Test Cohort)

Vehicle	Type	DPF Events (actual)	ML Alerts Triggered	Correct Alerts	False Alarms	Lead Time (hrs)
V35	Light truck	5	5	5	0	51.2
V36	Light truck	6	7	6	1	44.8
V37	Light truck	4	4	4	0	49.3
V38	Light truck	5	5	5	0	46.7
V39	Heavy haulage	7	7	7	0	48.1
V40	Heavy haulage	6	6	6	0	50.4
V41	Heavy haulage	5	6	5	1	43.9
V42	Tractor	5	5	5	0	45.6
TOTAL	—	43	45	43	2	47.3 ± 8.2

Table S18. Per-vehicle DPF alert performance on 8-vehicle test cohort.

S10. Comparison with Published Benchmarks

The table below directly compares key quantitative results from this paper against the most closely related published studies cited in the main paper.

Study	Task	Best Model	Accuracy / F1	Notes
This paper	DPF clog prediction	CNN-LSTM	94.3% / 0.937	42 vehicles, 18 months, India
Gao et al. [8]	DPF state estimation	Bayesian NN	89.7% / 0.891	Lab controlled; fewer vehicles
This paper	Fuel filter classification	XGBoost	91.8% / 0.914	6-class, commercial fleet
Wang et al. [9]	Fuel contamination classif.	XGBoost	93.1% / 0.928	4-class (simpler taxonomy)
Theissler et al.[6]	Multi-class auto fault diag.	Random Forest	82–88%	Broader fault scope, ECU data
Zhao et al. [7]	Component wear prediction	LSTM	87.3% / 0.869	Vibration sensors; single component

Table S19. Comparison with published benchmark results.

The small accuracy gap between this paper's XGBoost fuel filter result (91.8%) and Wang et al.'s (93.1%) is attributable to the more complex 6-class taxonomy used here versus 4 classes in Wang et al. A 4-class version of this study's model achieves 93.4%, directly comparable.

S11. System Architecture — Component Specifications

The following tables detail the hardware and software specifications for the proposed edge-cloud intelligent filter monitoring architecture.

S11.1 Edge Inference Node Specifications

Specification	Detail
Processor	ARM Cortex-M7 @ 216 MHz (STM32H743 series)
RAM	256 KB SRAM (on-chip)
Flash storage	2 MB on-chip + 8 MB external SPI flash
DPF model (deployed)	Pruned LSTM, INT8 quantized, ~38 KB
Fuel model (deployed)	INT8-quantized XGBoost, ~12 KB
Inference interval	1 minute
Inference latency	< 80 ms per cycle (both models combined)
Power consumption	< 180 mW during inference
Connectivity	4G LTE (primary), BLE 5.0 (local diagnostics)
Operating temperature	−40 °C to +85 °C
Quantization method	Post-training INT8 quantization (TensorFlow Lite)
Accuracy loss (quant.)	< 0.8% absolute accuracy vs full-precision model

Table S20. Edge Inference Node hardware and model specifications.

S11.2 Alert Threshold Parameters

Alert Level	DPF Health Score Threshold	Consecutive Cycles Required	Recommended Action
Level 0 (Normal)	> 0.35	—	No action
Level 1 (Warning)	0.15 – 0.35	3	Schedule forced active regeneration within 24 hrs
Level 2 (Critical)	< 0.15	1	Immediate workshop inspection
Fuel Fault Alert	P(fault) > 0.70	1	Fault-specific action from decision tree

Table S21. Alert threshold parameters for the Maintenance Decision Interface.

S12. Ethical Approval and Data Governance

Item	Detail
Ethics committee	Punjab Technical University Institutional Ethics Committee
Approval protocol ID	PTU-MECH-20220088
Scope of approval	Collection, storage, and analysis of vehicle telematics data for academic research
Data sharing agreements	Signed data-sharing agreements with all three fleet operators prior to data collection
Driver consent	All drivers informed of data collection; opt-out provision offered (0 opt-outs)
Data anonymisation	Vehicle IDs replaced with anonymised codes (V01–V42); no driver identifiers retained
Data retention	Raw sensor logs retained for 24 months from collection date, then deleted per agreement
Compute resources	HPC facility, Punjab Technical University (Grant PTU-HPC-2023-114)
Conflict of interest	None declared. No funding from vehicle manufacturers or filter suppliers.

Table S22. Ethical approval and data governance summary.

S13. References for Supplementary Claims

References below are cited within this supplementary document in addition to those cited in the main paper. Full main-paper references are listed in the main paper's reference section.

[S1] International Organization for Standardization, ISO 16750-1: Road vehicles — Environmental conditions and testing for electrical and electronic equipment, Geneva, Switzerland, 2023.

[S2] BOSCH Automotive Handbook, 10th ed., Stuttgart, Germany: Robert Bosch GmbH, 2022, Chapter 14: Exhaust-gas aftertreatment for diesel engines, pp. 892–947.

[S3] T. Chen and C. Guestrin, 'XGBoost: A scalable tree boosting system,' in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, San Francisco, CA, USA, pp. 785–794, 2016, doi: 10.1145/2939672.2939785.

[S4] F. Chollet, Deep Learning with Python, 2nd ed. Shelter Island, NY, USA: Manning Publications, 2021.

[S5] A. G. Konstandopoulos and J. H. Johnson, 'Wall-flow diesel particulate filters: Their pressure drop and collection efficiency,' SAE Technical Paper 890405, 1989, doi: 10.4271/890405.

[S6] Central Pollution Control Board (CPCB), 'Bharat Stage VI Emission Standards — Technical Specifications for Heavy-Duty Vehicles,' Ministry of Environment, Forest and Climate Change, Government of India, New Delhi, 2018.

[S7] B. McMahan et al., 'Communication-efficient learning of deep networks from decentralized data,' in Proc. 20th Int. Conf. Artificial Intelligence and Statistics (AISTATS), vol. 54, pp. 1273–1282, 2017.

[S8] TensorFlow Lite documentation, 'Post-training quantization,' Google LLC, https://www.tensorflow.org/lite/performance/post_training_quantization, accessed 2024.